

Piano: RNA-Seq Fusion Calling Pipeline Output

Introduction

The Piano pipeline (v2.3.3) is a wrapper around the Forte pipeline (v1.0.0, commit 0b3d927) for RNA-Seq fusion calling. It runs three independent fusion callers, merges and filters their predictions, annotates results against the OncoKB knowledge base, and produces a consolidated report.

The working version of Forte used here includes patch 01-forte-3071d5b-250807 (commit 20d39a1), which applies the upstream bugfix for per-core memory allocation (forte PR #162, bugfix/percore_mem_alloc).

Output Structure

The pipeline generates two main output directories:

```
{outdir}/
├── forte/          # Primary analysis results (per-sample and batch)
└── post/          # Post-processing results and final reports
```

Forte Directory

The forte/ directory (located under out/{ProjectID}/) contains all standard Forte pipeline outputs excluding BAM files. It is organized by sample:

```
forte/
├── analysis/
│   └── {SampleID}/
│       ├── arriba/          # Arriba fusion calls and visualizations
│       ├── fusioncatcher/   # FusionCatcher fusion calls
│       ├── starfusion/      # STAR-Fusion fusion calls
│       ├── metafusion/      # Merged/consolidated CFF files
│       ├── agfusion/        # Fusion protein domain visualizations
│       ├── agfusion_clinical/ # Visualizations for clinically relevant fusions
│       └── multiqc/          # Per-sample QC report
├── multiqc/                  # Batch-level QC report
├── pipeline_info/            # Nextflow execution metadata
└── runlog/                   # Input files and execution log
```

For a full description of the Forte outputs, refer to the Forte output documentation.

Post Directory

The post/ directory contains the consolidated fusion analysis report produced by Piano's post-processing scripts (runPost.sh). Post-processing performs two steps:

1. **OncoKB annotation** — unique gene-pair fusions are submitted to the OncoKB annotator (FusionAnnotator.py) to retrieve oncogenicity, mutation effect, and actionability levels.
2. **Report generation** — the R script reportFusion01.R reads the per-sample MetaFusion output (.final.cff files), merges OncoKB annotations, applies quality filters, and writes the final Excel and CSV reports.

Delivered Files

File	Description
{ProjectNo}__FusionTableV5.xlsx	Main fusion report workbook (two sheets)
{ProjectNo}__FusionTableV5__allEvents.csv	All detected fusion events (CSV)

An additional QC plot is generated in the working directory:

File	Description
qcCallerOverlap.pdf	UpSet plot showing overlap among the three fusion callers

FusionTable Excel Workbook

The workbook {ProjectNo}__FusionTableV5.xlsx contains two sheets.

HC.Events (High Confidence Events)

Fusion events that pass both of the following filters:

- Detected by **at least 2** of the 3 callers ($nCallers \geq 2$)
- Combined read support: $max_split_cnt + max_span_cnt \geq 5$

These represent the most reliable fusion predictions and are the recommended starting point for analysis.

AllEvents

The complete set of all detected fusion events, including single-caller predictions. This is the full unfiltered table and is useful for exploratory analysis or when investigating fusions in specific genes of interest. This sheet is also exported as the CSV file {ProjectNo}__FusionTableV5__allEvents.csv.

Column Descriptions

Column	Description
Sample	Sample identifier
Fusion	Fusion gene pair in the form Gene5--Gene3
Junction	Genomic breakpoint coordinates (chr:pos:strand) for both partners
max_split_cnt	Maximum split-read count supporting the junction across callers
max_span_cnt	Maximum spanning-read count supporting the junction across callers
FusionType	Structural class of the fusion (e.g., INTERCHROMOSOMAL, INTRACHROMOSOMAL, READ_THROUGH)
Fusion_effect	Predicted effect on the reading frame (e.g., in-frame, out-of-frame, frameshift)
Tools	Semicolon-delimited list of callers that detected this event
nCallers	Number of callers that detected this event (1, 2, or 3)
gene5_seq	Junction sequence from the 5' gene partner
gene3_seq	Junction sequence from the 3' gene partner
closest_exon5	Nearest exon boundary for the 5' gene partner
closest_exon3	Nearest exon boundary for the 3' gene partner
MUTATION_EFFECT	OncoKB mutation effect classification (e.g., Gain-of-function, Loss-of-function)
ONCOGENIC	OncoKB oncogenicity classification (e.g., Oncogenic, Likely Oncogenic, Unknown)
HIGHEST_LEVEL	Highest OncoKB therapeutic actionability level (Levels 1-4, R1-R2)
HIGHEST_SENSITIVE_LEVEL	Highest level of sensitivity to a therapeutic agent
HIGHEST_RESISTANCE_LEVEL	Highest level of resistance to a therapeutic agent
HIGHEST_DX_LEVEL	Highest OncoKB diagnostic actionability level (Dx1-Dx3)
HIGHEST_PX_LEVEL	Highest OncoKB prognostic actionability level (Px1-Px3)

OncoKB Actionability Levels

OncoKB levels reflect the strength of clinical evidence supporting the annotation:

- Level 1: FDA-recognized biomarker predictive of response to an FDA-approved drug
- Level 2: Standard care biomarker recommended by professional guidelines
- Level 3A/3B: Clinical evidence from trials or other tumor types
- Level 4: Compelling biological evidence
- R1/R2: Standard care or investigational resistance biomarker
- Dx1-Dx3: Diagnostic levels (1 = FDA/guideline-approved)
- Px1-Px3: Prognostic levels (1 = FDA/guideline-approved)

Empty cells in OncoKB columns indicate no annotation is available in the current OncoKB database for that gene pair.